

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276348

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

BARNES; Jeffery R. et al.

MODULAR DUAL BELT DRIVE UNIT

Abstract

Disclosed are lance hose drive units having a modular construction for easy disassembly and reassembly of its housing and components. A drive unit includes a housing having front and rear walls, and a top cover coupled to the walls. The housing may include a bottom cover removably also coupled to the front and rear walls, and a motor-side cover coupled to a side edge of at least one of the front wall, rear wall, and top cover. A drive unit may also comprise a belt drive unit positioned within the housing, and configured to drive movement of a lance hose positioned within the housing. A belt-side cover is removably coupled to the top cover, at least one toolless fastener removably securing the belt-side cover. The belt drive unit is slidably coupled at its front end within an interior front channel and at its rear end within an interior rear channel.

Inventors: BARNES; Jeffery R. (Ignacio, CO), KRAUSER; John L. (Durango, CO)

Applicant: STONEAGE, INC. (Durango, CO)

Family ID: 96881582

Appl. No.: 19/065511

Filed: February 27, 2025

Related U.S. Application Data

us-provisional-application US 63560914 20240304

Publication Classification

Int. Cl.: B08B9/043 (20060101); B65H51/10 (20060101)

U.S. Cl.:

CPC B08B9/0433 (20130101); B65H51/105 (20130101); B65H2701/33 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Application No. 63/560,914, filed Mar. 4, 2024, entitled “Modular Dual Belt Drive Unit,” the entirety of which is hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure is generally related to the field of high pressure fluid rotary nozzle handling systems, and more particularly to a novel modular nozzle handling unit for use with such handling systems, as well as methods of use and methods of manufacture thereof.

BACKGROUND

[0003] High pressure fluid rotary nozzle handling systems are employed for advancing and retracting one or more flexible tube cleaning lances from tubes arranged in an array, such as in a heat exchanger, from a position adjacent a heat exchanger tube sheet. Advanced systems further include automatically, repetitively reversing forward lance hose feed movement upon encountering an obstruction within a tube or other piping system being cleaned.

[0004] One conventional tube lancing unit consists of a rotating reel flexible lance hose take-up and hose dispensing apparatus that carries a predetermined length of flexible lance hose wrapped around a drum. The reel in the drum is rotated by an air motor to push the flexible lance out of the drum and into one or two heat exchanger tubes. The air motor drive can be automatically reversed upon pneumatically sensing a large air pressure increase in air pressure supplied to the forward directional side motor that occurs if the flexible lance being pushed by the reel rotation encounters an obstruction within a tube being cleaned.

[0005] As with many types and models of lance hose drive units, failures can occur during use. For example, one or more belt drive units used to move the lance hose(s) through the drive unit may fail, such as the endless belt failing, one or more rollers on a belt drive unit failing, or perhaps the motor driving a belt drive unit failing. Furthermore, a hose being driven by and through the drive unit may become jammed or even broken within the drive unit. Thus, when such a failure occurs, the drive unit must be stopped and taken out of use for service or repair. Additionally, drive units may need to be cleaned, further requiring them to be taken out of use for cleaning. Typically, repairing, servicing, or cleaning a drive unit requires disassembling the drive unit, which traditionally is a long and arduous process. Fasteners such as screws must be removed to open and disassemble the drive unit, as well as to remove and replace or repair components within the drive unit. Doing so is a time-consuming process and requires taking the drive unit out of use for an undesirable amount of time. In some situations, a drive unit in need of repair may even need to be sent to repair facility for disassembly and repair, further increasing the length of time such a drive unit is out of use.

[0006] Additionally, components within a drive unit needing repair, cleaning, maintenance, or replacement are typically similarly difficult to remove from the drive unit. Again, such difficult removal of internal components of a drive unit may again cause undesirable delay and downtime for the drive unit. Therefore, what is needed in the art is a hose drive unit and related methods for quickly disassembling and reassembling the drive unit when needed in order to reduce downtime and increase ease of repair or replacement of components within the drive unit. The disclosed principles provide for such a unit and related methods.

SUMMARY

[0007] The disclosed principles provide embodiments of a flexible high pressure fluid cleaning lance tractor drive apparatus or unit having a modular design and construction for quick and easy disassembly and reassembly of the drive unit, its housing, and the components therein. In one embodiment, a drive unit in accordance with the disclosed principles may comprise a housing

having a front wall, a rear wall, and a top cover coupled to top ends of the front and rear walls. The housing may also include a bottom cover removably coupled to bottom ends of the front and rear walls, as well as a motor-side cover coupled to a side edge of at least one of the front wall, rear wall, and top cover. The exemplary drive unit may also comprise upper and lower belt drive units positioned within the housing, and configured to drive movement of at least one lance hose positioned between the belt drive units. An actuator is coupled between the top cover and the upper belt drive unit, and configured to provide vertical movement of the upper belt drive unit. A belt-side cover is removably coupled to the upper belt drive unit via at least one fastener, where the belt-side cover is configured to move vertically with the upper belt drive unit via the actuator. In such drive units, the upper and lower belt drive units are slidably coupled at their front ends within a channel formed on an interior surface of the front wall, and at their rear ends within a channel formed on an interior surface of the rear wall, and the bottom cover may also be removably coupled to a bottom edge of the lower belt drive unit providing structural support therefor. Fasteners may be included at lower portions of the front and rear walls sized and configured to retain the belt drive units within the channels on the front and rear walls.

[0008] Additional embodiments and advantages and variation thereof are also encompassed within the scope of the disclosed principles, and some such exemplary embodiments discussed in further detail herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The novel features believed characteristic of the disclosure are set forth in the appended claims. The disclosure itself, however, as well as a preferred mode of use, further objectives, and advantages thereof, will be best understood by reference to the following detailed description of illustrative embodiments when read in conjunction with the accompanying drawing, in which:

[0010] FIG. 1 illustrates a perspective view of a conventional flexible lance hose drive system that may include a drive unit designed and constructed in accordance with the disclosed principles;

[0011] FIG. 2 illustrates an exemplary embodiment of a multiple lance drive unit in accordance with the disclosed principles;

[0012] FIG. 3A illustrates an isometric view of the drive unit of FIG. 2, but viewed from an opposite direction, with the motor-side cover installed;

[0013] FIG. 3B illustrates an isometric view of the drive unit of FIG. 3A with the motor-side cover removed;

[0014] FIG. 4 illustrates a belt-side view of the drive unit shown in FIGS. 2, 3A and 3B with its upper and lower belt-side covers removed;

[0015] FIG. 5 illustrates a partially exploded view of the drive unit shown in FIGS. 2, 3A and 3B, with the upper and lower belt drive units slid out of the housing;

[0016] FIG. 6 illustrates an alternative partially exploded view of the drive unit shown in FIGS. 2, 3A and 3B;

[0017] FIG. 7 illustrates a sectional view of a rear portion of a drive unit in accordance with the disclosed principles;

[0018] FIG. 8 illustrates a sectional view of a bottom portion of the drive unit illustrated in FIG. 7; and

[0019] FIG. 9 illustrates a partially transparent view of the belt-side of the drive unit illustrated in FIG. 8.

DETAILED DESCRIPTION

[0020] For the purpose of promoting an understanding of the principles of the invention, reference will now be made to the embodiments illustrated in the drawings and specific language will be

used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Any alterations and further modifications in the described embodiments, and any further applications of the principles of the invention as described herein are contemplated as would normally occur to one skilled in the art to which the invention relates. Although multiple embodiments are shown and discussed in great detail, it will be apparent to those skilled in the relevant art that some features that are not relevant to the present invention may not be shown for the sake of clarity.

[0021] As used in this disclosure, the term “rear” refers the inlet end of the drive unit configured to receive one or more flexible lances, and the term “front” refers the outlet end of the drive unit configured to discharge the one or more flexible lances.

[0022] Turning initially to FIG. 1, illustrated is a perspective view of a conventional flexible lance hose drive system **100**, and which may include a novel drive unit or apparatus designed and constructed in accordance with the disclosed principles. In this embodiment, the hose drive system **100** includes a hose drive unit **102** and a hose guide assembly **104** coupled to an output end of the hose drive unit **102**. The hose drive unit **102** may be configured to drive any number of flexible lances **101**, each comprising a lance hose **103** coupled to a nozzle **105**. The drive unit **102** may be a one, two, or three lance drive like the illustrated embodiment. The hose drive unit **102** may be replaced with a modular drive unit in accordance with the disclosed principles, while the remainder of the system **100** remains unchanged. As such, a modular unit as disclosed herein can be substituted into existing systems without a need to alter or otherwise adapt existing system components.

[0023] The guide assembly **104** includes, in this exemplary embodiment **100**, a set of three guide tubes **108** adjustably fastened to a bracket **110** fastened to a trolley (not illustrated) along with a sensor amplifier block **112** beneath the tubes **108** and fastened to the bracket **110**. The tractor drive unit **102** is fastened to the bracket **110** via a hose stop collet or crimp encoder block **114** fastened to a rear end of the set of three guide tubes **108**. Each of the guide tubes **108** is an elongated cylindrical tube, preferably made of a metal, such as stainless steel, aluminum, brass, a durable plastic, or other rigid material with a high electrical resistivity. An AC pulse sensor **116** is mounted at the distal end of each guide tube **108**.

[0024] Turning to FIG. 2, illustrated is an exemplary embodiment of a multiple lance, dual belt drive unit **200** in accordance with the disclosed principles. The drive unit **200** is configured to drive flexible high pressure fluid cleaning lance hoses, such as those described with reference to FIG. 1. A belt-side view of the drive unit **200** is shown in FIG. 2 with its removable belt-side covers **230** and **240** in place, while a perspective view of the drive unit **200** with upper and lower outer belt-side covers **230** and **240** removed is shown in FIG. 4. The drive unit **200** has a rectangular box housing that includes, in addition to the belt-side covers **230**, **240**, a flat top plate or cover **204**, a flat bottom plate or cover **206**, front and rear walls **208** and **210**, respectively, a motor-side cover (see FIG. 3), and two carry handles **212**, one on each of the front and rear walls **208** and **210**.

[0025] Fastened to the front wall **208** is an exit hose guide manifold **214**. Fastened to the rear wall **210** below the carry handle **212** is a hose entrance guide manifold **218**. Either or both these manifolds **214** and **218** may include a set of hose guide collets for guiding one to three flexible lance hoses (not shown) into and/or out of the housing of the drive unit **200**, and onto the belt drive units housed therein and discussed below. Each guide collet set is sized to accommodate a particular lance hose diameter. Hence, the collet sets are changeable depending on the lance hose size to be driven by the drive unit **200**. Toolless fasteners **214A** and **218A** may be provided to secure accessories to respective manifolds **214**, **218** to the drive unit **200**. Toolless fastener **214B** is provided to permit removal/replacement of the collets in the exit guide manifold **214**. Permitting such quick, toolless replacement of the collets permits the size of lances used with the drive unit **200** to be easily changed in the field with very little downtime for the unit **200**.

[0026] Each of the manifolds **214** and **218** includes a sensor, typically a Hall effect sensor (not

shown), for detecting the presence or absence of a metal hose stop element that is fastened to each flexible lance hose. These sensors are used to stop the belt drive units in the drive unit **200** when the presence of a hose stop element is sensed. One hose stop element is preferably integrated into the threaded hose ferrule to which a nozzle is attached, at the end of each of the lance hoses. This particular type of hose stop element is configured to prevent inadvertent withdrawal of the flexible lance hose out of the heat exchanger tube sheet and into the drive unit **200**. Exemplary sensors and related detection techniques may be seen in U.S. Pat. Nos. 11,441,856 and 11,781,852, which are co-owned with the present disclosure and incorporated herein by reference in their entireties as if reproduced herein. The collection of the entrance guide manifold, corresponding guide collets, and any sensor(s) comprise a hose entrance assembly, and the collection of the exit guide manifold, corresponding guide collets, and any sensor(s) comprise a hose exit assembly.

[0027] Referring briefly to FIG. 3A, illustrated is an isometric view of the drive unit **200** of FIG. 2, but viewed from an opposite direction. FIG. 3B illustrates an isometric view of the drive unit **200** with the motor-side cover **225** removed. Toolless fasteners, for example, threaded fasteners **225A**, may be included on motor-side cover **225** for permitting toolless removal of the motor-side cover **225**. In some embodiments, magnetic fasteners may be used in place of threaded fasteners **225A**, while in other embodiments, quarter-turn fasteners (such as fasteners **265A**, **265B** discussed below and illustrated in FIG. 7) may instead be employed. Motor-side cover **225** is illustrated in the uninstalled position, and removal of this cover **225** permits access to one or more motors **226**, **228**, as well as access to one or more fasteners **230A** (see FIG. 4).

[0028] Looking now at FIG. 4, illustrated is a belt-side view of the drive unit **200** shown in FIGS. 2 and 3 with its upper and lower belt-side covers **230**, **240** removed. The upper and lower belt-side covers **230**, **240** are designed and constructed so as to structurally connect together when installed on the drive unit **200** (specifically, a bottom edge of the upper cover **230** is configured to structurally but removably engage a top edge of the lower cover **240**), and the upper cover **230** is removably affixed to the drive unit **200** via cover mounting pins **230A**. The mounting pins **230A** may also be employed to structurally connect the upper belt drive unit **222** to the clamping mechanism **204A**, which as described in detail below is used to vertically move the upper belt drive unit **222** within the unit **200** to clamp lance hoses positioned and fed between the upper and lower belt drive units **222**, **224**.

[0029] In exemplary embodiments, the top cover **204** is secured to the top ends of both the front and rear walls **208**, **210** with fasteners requiring tools, such as screws. Additionally, internal structural components may similarly be secured to these components with such tool-requiring fasteners. The securing of the top cover **204** to the front and rear walls **208**, **210**, as well as any internal structure, in this manner provides a sturdy base structure for the drive unit **200** that is maintained while other components are configured as removable from this base structure in the quick-disconnect, tool-free manner disclosed herein. Pins or other fastening devices may be removably connected to ends of the mounting pins **230A** to secure the upper belt-side cover **230** to the drive unit **200**. Alternatively, ends of the pins **230A** may include fastening devices, such as spring-loaded tabs, pins, or similar toolless device that removably engage and release the upper cover **230**.

[0030] The drive unit **200** includes an inner vertical support wall **220** removably fastened to the front and rear walls **208** and **210**, and the top and bottom plates **204** and **206**. This vertical support wall **220** is part of the upper and lower belt drive units **222**, **224**, and may be part of, or comprise the entirety of, the internal vertical support structure in embodiments of the drive unit **200** having such a structure. This vertical support wall **220** divides the housing of the drive unit **200** and the belt drive units **222**, **224** into a belt-side portion and a motor-side portion. The motor-side portion primarily houses hose fittings and belt drive motors **226** and **228**, along with any needed wiring, controllers, etc. The belt-side portion is a belt cavity through which flexible lance hoses (not shown) are driven by the endless belts disposed on the upper and lower drive belt units **222** and

224, respectively, when fed therebetween.

[0031] In this exemplary embodiment of drive unit **200**, the inner vertical support wall **220** is shown to be formed in two interconnected parts, an upper part **220A** and a lower part **220B** (collectively **220**), as seen in FIG. 5. FIG. 5 illustrates a partially exploded view of the drive unit **200**, with the upper and lower belt drive units **222**, **224** slid out of the housing. A separate upper support wall **220A** aligned with and removably connected to the lower support wall **220B** provides a rigid, structural joint between the front and rear walls **208** and **210**, as well as the top plate **204**, and optionally in some embodiments with the bottom plate **206** as well. The structural connection between the upper and lower support walls **220A**, **220B** and various components of the housing is described in further detail below. Each of the support walls **220A**, **220B** is preferably a flat plate of a lightweight material such as aluminum, or could be made of a structural polymer with sufficient strength and rigidity to handle the motor operational stresses involved.

[0032] Each of the support walls **220A**, **220B** carries a drive motor **226** and **228**, respectively, mounted such that their drive shafts **226A**, **228A** protrude laterally through each of the support walls **220A**, **220B** into the belt-side portion and into a drive end of the upper and lower drive belt units **222**, **224**. Each of the drive motors **226** and **228** has a dominant length dimension that is aligned in parallel with the belts of the belt drive units **222**, **224**, and thus each includes a gearbox **226A**, **228A** to transfer and redirect the rotation of the drive shaft of each motor into drive rollers **226B**, **228B** onto which belts of the belt drive units **222**, **224** are mounted. Such parallel alignment of the motors **226**, **228** with the belt drive units **222**, **224** permits a more compact overall construction of the drive unit **200** and allows each of the drive motors **226** and **228** to be housed entirely within the housing of the drive unit **200**. Each of the drive motors **226**, **228** is connected to pneumatic forward feed line **234** and reverse feed line **232** (see FIG. 2) through a feed manifold fastened to the rear plate **210**. Each of the drive motors **226** and **228** shown in FIG. 5 is preferably a compact vane gearmotor with integral planetary gear reduction driving a bevel gearbox.

Alternatively, radial piston pneumatic, hydraulic or electric motors could be used.

[0033] Importantly, the distal ends of each of the drive shafts **226A** and **228A** terminate before reaching the belt-side support covers **230**, **240**, and those drive shafts are laterally supported from the respective upper and lower support plates **220A**, **220B**. Such centralized support within the interior of the drive unit **200** permits the outer side walls of the drive unit **200** to be removed without affecting the structural integrity of the drive belt units **222**, **224** and their respective drive motors **226**, **228**. A separate bearing supported shaft also extends out of, and is supported by, respective upper and lower support plates **220A**, **220B**, and on which each roller of the drive belts **222**, **224** is supported. For example, spaced between each splined drive roller **226B**, **228B** at the drive end of each of the drive belt units **222**, **224**, and corresponding splined idler rollers **226C**, **228C** opposite the drive rollers, are cylindrical guide rollers (not visible) that are also supported by the respective upper and lower support plates **220A**, **220B** via shafts extending therefrom and terminating in outer side plates **226D**, **228D** of each belt drive unit **222**, **224**. The outer surface portion or cover of the endless belts on the drive belt units **222**, **224** may be flat and smooth to prevent undesirable hose abrasion and degradation as they drive lance hoses between them and are formed of a suitable friction material such as polyurethane. In other embodiments, the belts may have circumferential grooves (e.g., 3 grooves if 3 hoses are to be driven) to stabilize the hoses while contacting the belts to resist buckling of smaller hose sizes. Such belts may also have perpendicular grooves (sipes) on, for example, a 10 mm pitch to help shed debris from the outer diameters of the hoses during use. The idler rollers **226C**, **228C** may be slid forward or backward to tighten or loosen, respectively, the endless belts of the belt drive units **222**, **224**. Tensioner pins **226E**, **228E** may be provided to tighten or release the idler rollers **226C**, **228C** in order to tighten or loosen the belts. These tensioner pins **226E**, **228E** may again be toolless so as to facilitate quick field disassembly of the unit **200**, and in particular replacement of the belts on the belt drive units **222**, **224**, in the manner described herein.

[0034] A vertical actuator, which in this embodiment is pneumatic clamping mechanism **204A**, attaches to the top plate **204** with fasteners, such as flathead screws, and attaches to the upper belt drive assembly **222** with pins **230A**, as described above. The clamping mechanism **204A** moves the upper belt drive assembly **222** in relation to the lower belt drive assembly **224** to allow multiple lance hose sizes to be used with drive unit **200**. Adjusting the clamping pressure provided by the clamping mechanism **204A** also controls the amount of clamping force on the hoses and the resulting traction transmitted from the belts on the upper and lower belt drive units **222**, **224** to the cover of the hose(s) employed with the drive unit **200**. The clamping mechanism **204A** is described in further detail below with reference to FIG. 9.

[0035] Also illustrated in FIG. 5 is a component of the interior support structure formed vertically along an interior of the rear wall **210**. In this embodiment, this component of the support structure is a channel **245B** formed vertically along a substantial length of the rear wall **210**. The interior of the front wall **208** includes a corresponding channel **245A** disposed thereon, and partially visible in FIG. 4. The front wall channel **245A** is sized to slidably receive the front edges of the upper and lower support walls **220** therein, while the rear wall channel **245B** is sized to slidably receive the rear edges of the upper and lower support walls **220** therein. Moreover, the channels **245A**, **245B** are sized such that, despite permitting a slidable connection with the front and rear edges, respectively, of the upper and lower support walls **220**, the placement of the support walls **220** within the channels **245A**, **245B**, as well as their connection to one another, still provides structural integrity within the housing, and thus the drive unit **200** as a whole. This is important since, as discussed above, the belt drives **222**, **224** are carried on and structurally supported by the support walls **220**, and thus the channels **245A**, **245B** assist in proving structural integrity between the belt drive units **222**, **224** and the other components of the housing, in particular the top cover **204** and the front and rear walls **208**, **210** adjoined thereto with tooled fasteners.

[0036] The top edge of the upper support wall **220A** may also removably engage with one or more structures formed on an interior surface of the top cover **204**. In this embodiment, one or both of toolless fasteners **230A** may be used to secure the upper support wall **220A** via one or more corresponding apertures formed therethrough to an interior of the top of the housing. This in turn secures the upper belt drive unit **222** within the drive unit **200**. Removal of the motor-side cover **225** can permit access to fasteners **230A**, which may be pulled by handles **230B** to release the upper belt drive unit **222** and to optionally remove the belt side cover **230**.

[0037] Additionally, a bottom edge of the lower support wall **220B** may also removably engage with one or more structures formed on an interior surface of the bottom cover **206** to further provide structure to the drive unit **200**. In particular, in some embodiments, the interior surface of bottom cover **206** includes a channel **206A** (shown in FIG. 7) formed therein or thereon and extending longitudinally at least partially between its front end to its rear end. This channel **206A** may be sized to removably receive a bottom edge of the lower support wall **220B** therein, and is sized such that, despite permitting a slidable connection with the bottom edge of the lower support wall **220B** and the bottom cover **206**, the placement of the lower support wall **220B** within the channel **206A** still provides structural integrity within the housing, and thus the drive unit **200** as a whole. Specifically, structural integrity is provided between top and bottom covers **204**, **206** and the upper and lower walls **220A**, **220B**.

[0038] Alternatively or as part of the channel **206A**, two end channels **206B**, with one formed at each of the front and rear ends of the bottom cover **206**, may instead be sized to receive first and second toolless fasteners **206C** respectively located through bottom ends of the front and rear walls **208**, **210**. In exemplary embodiments, one of these fasteners **206C** passes through a bottom end of the front wall **208** at a terminal end of the channel **245A** located on the interior of the front wall **208**, and the other of these fasteners **206C** passes through a bottom end of the rear wall **210** at a terminal end of the channel **245B** on the rear wall **210**. These toolless fasteners **206C** are positioned in the front and rear walls at these locations to prevent sliding withdrawal of the lower support wall

220B of the lower belt drive unit **224** from within the channels **245A**, **245B** on the front and rear walls respectively. In such embodiments, the channels **206B** in the bottom cover **206** are formed to provide room for the fasteners **206C** when the bottom cover **206** is affixed to the drive unit **200**. As with other toolless fasteners disclosed herein, fasteners **206C** may include handles with which to pull the fasteners **206C** from within corresponding apertures formed in the front and rear walls **208**, **210** to permit sliding out of the upper and lower support walls **220** from the front and rear channels **245A**, **245B**. However, in some embodiments, the fasteners **206B** may also be spring-loaded so that when pulled, the fasteners **206B** permit sliding out of the upper and lower support walls **220** from the front and rear channels **245A**, **245B**, and then they return to their original position when the handles are released.

[0039] Additional toolless fasteners providing removable coupling of various components of the drive unit **200** may also be included to permit the quick, toolless disassembly of operative components of the drive unit **200** as disclosed herein. For example, illustrated in FIGS. 2, 3 and 4 are pull pins **208A** configured to pass through corresponding apertures formed through a bottom portion of the front wall **208** (see FIGS. 3 and 4), as well as pull pins **210A** configured to pass through corresponding apertures formed through a bottom portion of the rear wall **210** (see FIGS. 2 and 3). Pull pins **210A** are sized to be removably received through apertures **210B** formed in a rear edge of the bottom cover **206**, and pull pins **208A** are sized to be removably received through apertures **208B** formed in a front edge of the bottom cover **206**. As such, pull pins **208A** may be quickly removed without the need for any tool from their corresponding apertures formed in the front edge of the bottom cover **206**, and pull pins **210A** may be similarly removed from their corresponding apertures **210B** formed in the rear edge of the bottom cover **206**, in order to quickly remove the bottom cover **206**. Alternatively, as in other embodiments disclosed herein, the pull pins **208A**, **210A** may again be spring-loaded so that when pulled, the distal ends of pull pins **208A** are removed from the apertures on the front edge of the bottom cover **206**, but then remain connected to their respective apertures in the front wall **208**. Likewise, in such embodiments, the distal ends of pull pins **210A** may be removed from the apertures on the rear edge of the bottom cover **206**, but then remain connected to their respective apertures in the rear wall **210**.

[0040] Once the bottom cover **206** is removed via removal of the pull pins **208A**, **210A**, each of the pull pins **208A**, **210A** may then return to their original position until the bottom cover **206** is to be reattached as part of the housing of the drive unit **200**. Then, the pull pins **208A**, **210A** are again pulled until the apertures in the front and rear edges of the bottom cover **206** are aligned with the distal ends of the pull pins **208A**, **210A**, and the pins reinserted into those apertures. In embodiments where the pull pins **208A**, **210A** are simply removed from the front and rear walls **208**, **210**, the respective pull pins **208A**, **210A** are simply passed back through their corresponding apertures formed at the bottom of the front and rear walls **208**, **210**, and then into the apertures formed in the front and rear edges of the bottom cover **206**. In exemplary embodiments, the distal ends of the pins **208A**, **210A** are sized for clearance fit within the corresponding apertures in the front and rear edges of the bottom cover **206**. Although only two pull pins **208A**, **210A** are illustrated for securing each of the respective front and rear edges of the bottom cover **206**, in other embodiments a greater or fewer number of pull pins may be included.

[0041] Turning now to FIG. 6, illustrated is an alternative partially exploded view of the drive unit **200** of FIG. 5. Once again, the upper and lower belt drive units **222**, **224** are shown removed from the housing of the drive unit **200**. From this view, the channel **245A** formed along a length of the interior of the front wall **208** is also visible. As discussed in detail above, this channel **245A** and the channel **245B** formed along a length of the interior of the rear wall **210** are sized and aligned to slidably receive therein the front and rear edges of the upper and lower support walls **220A**, **220B**. As the support walls **220A**, **220B** are slid into or out of the drive unit **200**, the belt drive units **222**, **224** mounted respectively thereon are also movable into and out of the drive unit **200** quickly and easily. As such, one or both of the belt drive units **222**, **224** may then be serviced, cleaned, repaired,

or replaced, as needed.

[0042] Upper toolless fasteners **250A** are also visible in the illustration in FIG. **6**. These fasteners **250A** may be employed in the housing of the drive unit **200** to facilitate securing and removal of the upper belt-side cover **230** of the drive unit **200**. Lower toolless fasteners **255A** are also visible, and may be used to facilitate securing and removal of the lower belt-side cover **240** from the drive unit **200**. To employ the fasteners **250A**, **255A** to remove these covers **230**, **240** of the drive unit **200** housing, the fasteners **250A**, **255A** may be threaded and thus loosened or altogether removed by unthreading them from their corresponding apertures formed in the front and rear walls **208**, **210**, and then slidable locks **250** may be moved from a locked to an unlocked position.

Alternatively, the fasteners **250A**, **255A** may be slidably connected to the front and rear walls **208**, **210**, and thus the fasteners **250A**, **255A** may be loosened and slid along with the locks **250** from the locked to unlocked position so that the appropriate cover **230**, **240** secured by the locks **250** may be removed. Still further, the fasteners **250A**, **255A** may be pins, such as spring-loaded pins, that may be operated to facilitate removal of the upper and lower covers **230**, **240**.

[0043] Looking back briefly to FIG. **4**, notches **230C** and corresponding tabs **230B** formed on front and rear edges of the upper belt-side cover **230** may be sized to correspond to the position of cams **250C** positioned on the slidable locks **250**. The alignment of these cams **250C** with notches **230C** at the upper portion of the upper belt-side cover **230** when the fasteners **250A** are loosened so that the slidable locks **250** (see FIGS. **6** and **7**) are slid to their unlocked position to permit removal of the cover **230** from the drive unit **200**, while the position of the cams **250C** when the fasteners **250A** tightened once the slidable locks **250** are slid to their locked position can secure the upper cover **230** via tabs **230B** (adjacent the notches **230C**) as part of the housing. Sliding movement of the slidable locks **250** and upper fasteners **250A** between locked and unlocked positions may be configured to solely move the cams **250C** from alignment with a notch **230C** or tab **230B** at the top edge of the upper cover **230**. In such embodiments, when the slidable locks **250**, and thus the cams **250C**, are slid downwardly to the unlocked position, and thus in alignment with a notch **230C** proximate the top of the upper cover **230**, the cover **230** may be slid upward until additional notches formed along the front and rear edges of the cover **230** are aligned with corresponding additional cams **250C** such that the cover **230** may then be laterally removed from the drive unit **200** without the need for any tools. Similarly, the lower fasteners **255A** may also include be tightened or loosened to facilitate sliding of the slidable locks **250** (and thus lower cams **255C**) where alignment of these lower cams **255C** with notches at the lower portion of the lower belt-side cover **240** when the slidable locks **250** are slid to their unlocked position can permit removal of the lower cover **240** from the drive unit **200**, while the position of these lower cams **255C** when the locks **250** are slid to their locked position can secure the lower cover **240** as part of the housing to retain the lower cover **240** by positioning tabs adjacent to the notches behind the lower cams **255C**. While in this embodiment, single slidable locks **250** carry both cams **250C** as well as the lower cams **255C** such that both the upper and lower covers **230**, **240** can be removed simultaneously. In some embodiments, separate slidable locks may be provided so that the covers **230**, **240** may be removed separately.

[0044] Returning to FIG. **6**, toolless lock pull pins **250B** may be provided and secured within corresponding apertures to further prevent sliding of the slidable locks **250** when in the locked position, and then when removed, permit sliding of the locks **250** to an unlocked position to permit removal of the appropriate cover **230**. As before, the pull pins **250B** may also be spring-loaded so that when pulled, the locks **250** may be slid to their unlocked position to permit removal of the upper and lower belt-side covers **230**, **240** from the drive unit **200**, and then the pins **250B** return to their original position when their handles are released. Alternatively, the pins **250B** may be pull pins that are simply removed from their respective apertures to permit sliding of the locks **250**.

[0045] Referring now to FIG. **7**, illustrated is a sectional view of a rear portion of a drive unit **200** in accordance with the disclosed principles with the disclosed principles. This view of the drive

unit **200** further illustrates toolless securing pins **260A** employed to secure the upper belt-side cover **230** to the upper belt-drive unit **222**, as well as toolless securing pins **260B** employed to secure the lower belt-side cover **240** to the lower belt-drive unit **224**. In this illustrated embodiment of the toolless securing pins **260A**, **260B** (collectively **260**), the securing pins **260** are twistable pins that when twisted to a locked orientation, distal ends of the securing pins **260** are secured into a mating aperture within the belt drive units **222**, **224**. Likewise, when the securing pins **260** are twisted to an unlocked orientation, distal ends of the securing pins **260** are permitted to be released from their corresponding mating apertures, and thus the pins **260** removed from the belt drive units **222**, **224** again without requiring the use of any tool in order to facilitate quick field disassembly of the unit **200**.

[0046] In this embodiment, the mating apertures are formed within sides of the belt drive units **222**, **224**, but in other embodiments the securing pins **260** may be removably connected to another component or structure within the drive unit **200**. Looking closer at the securing pins **260**, their distal ends may include corresponding elongated tabs or pins **265A**, **265B** that are sized to pass into and out of corresponding slotted apertures to either release or secure the securing pins **260** in place, and thus secure or assist in releasing the upper and lower belt-side covers **230**, **240**. In this embodiment, the tabs **265A**, **265B** are configured such that a 90 degree rotation of the securing pins **260** defines the locked and unlocked orientations; however, other degrees of rotation may also be provided. Such securing pins **260** provide for quick and easy unlocking of the pins **260** from the drive unit **200**, thus permitting quick removal of the belt-side covers **230**, **240**. Furthermore, springs may be provided within the heads (or under the heads, as illustrated) of each of the securing pins **260** to bias the heads outwardly from the belt-side covers **230**, **240** when secured with the securing pins **260**. This ensures the securing pins **260** do not come loose during use of the drive unit **200**. In still other embodiments, the securing pins **260** may instead be threaded and thus configured to thread into corresponding threaded apertures, and thus the securing pins **260** are unthreaded to remove the covers **230**, **240**.

[0047] Also partially shown in FIG. 7 is the clamping mechanism **204A**. As mentioned above, the clamping mechanism **204A** is attached to the support wall **220A** and belt-side cover **230** of the upper belt drive **222**, and thus is employed to raise or lower the belt drive **222** to provide a clamping force to the lance hoses held between the upper and lower belt drive units **222**, **224**. The lower edge of the upper support wall **220A** and the upper edge of the lower support wall **220B** are configured to have a sliding relationship such that a structural relationship still exists between the upper and lower support walls **220A**, **220B** despite the vertical movement of the upper belt drive **222** with the clamping mechanism **204A**. Similarly, the lower edge of the upper belt-side cover **230** and the upper edge of the lower belt-side cover **240** may also be configured to have a sliding relationship such that a structural relationship remains between the two covers **230**, **240** throughout the range of vertical movement of the upper belt drive unit **222** provided by the clamping mechanism **204A**. Such a sliding relationship may also be provided between the front and rear edges of the upper belt-side cover **230** and grooves formed on the interiors of the front and rear walls **208**, **210** within the interior sidewalls of the slidable locks **250**. Thus, while a slidable movement of the front and rear edges (including the tabs **230B** and notches **230C**) is provided in such grooves, the upper and lower cams **250C**, **255C** still retain the upper and lower covers **230**, **240** to the unit **200** when the slidable locks **250** are maintained in their locked position (e.g., upward in this embodiment) and secured as such with the upper and lower fasteners **250A**, **255A**, and pins **250B**.

[0048] Looking now at FIG. 8, illustrated is a sectional view of a bottom portion of the drive unit **200** illustrated in FIG. 7. This sectional view further illustrates the lower set of securing pins **260B**, which are also illustrated in cross-section. Also shown are the elongated tabs or pins **265B** positioned near the distal ends of the securing pins **260B**. In addition, in some embodiments, the far distal ends of these securing pins **260A**, **260B** are positioned within apertures formed in the sides of

the belt drive units **222**, **224**, which assists in providing support for the belt drive units **222**, **224**. Specifically, the middle portion of the securing pins **260A**, **260B** may contact apertures formed in the belt-side covers **230**, **240**, while the distal ends of the securing pins **260A**, **260B** contact apertures, e.g., via clearance fit, in the belt drive units **222**, **224**, thereby providing a structural connection between the covers **230**, **240** and their respective belt drive units **222**, **224**.

[0049] Turning finally to FIG. **9**, illustrated is a partially transparent view of the belt-side of the drive unit **200**. Support members **275A**, **275B** are affixed to the tops of the front and rear walls **208**, **210**, and top plate **204**, of the drive unit **200**, and the actuator, in the form of clamping mechanism **204A**, is provided between these support members **275A**, **275B**. Support member **275B** is shown transparent in FIG. **9**, as is the near sidewall of the clamping mechanism **204A**, to permit easier viewing of the interior of the clamping mechanism **204A**.

[0050] As shown in this embodiment, the assembly of the clamping mechanism **204A** includes three pistons **270**, as well as position housing and cover surrounding the pistons and other internal components of the clamping mechanism. **204A**. Top ends **270A** of the pistons **270** may be secured to the top cover **204**, for example, using screws or similar fasteners. The piston housing and cover move vertically, for example, from a pneumatic driving force, in relation to the top plate **204**, thus moving the entire upper belt drive unit **222** vertically with respect to the lower belt drive unit **224**. Pins **230A** are positioned through the piston housing of the clamping mechanism **204A**, the upper cover **230**, and upper support plate **220A** as well. Retractions springs **270B** are provided around each of the pistons **270** to provide a return force during use of the clamping mechanism **204A**. As discussed above, the lower edge of the upper support plate **220A** maintains a slidable connection with the top edge of the lower support plate **220B**, and the bottom edge of the upper belt-side cover **230** maintains a slidable connection with the top edge of the lower belt-side cover **240**, and the upper belt drive unit **222** is vertically moved by the actuator **204A** during use of the drive unit **200**. While the clamping mechanism **204A** in this embodiment is shown as a pneumatically driven mechanism, other drive types are also possible. For example, the clamping mechanism **204A** may be hydraulically driven or electrically driven, as well as any other advantageous drive technology.

[0051] While this disclosure has been particularly shown and described with reference to preferred embodiments, it will be understood by those skilled in the pertinent field art that various changes in form and detail may be made therein without departing from the spirit and scope of the invention. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend the invention to be practiced otherwise than as specifically described herein. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto, as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

[0052] Also, while various embodiments in accordance with the principles disclosed herein have been described above, it should be understood that they have been presented by way of example only, and not limitation. Thus, the breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with any claims and their equivalents issuing from this disclosure. Furthermore, the above advantages and features are provided in described embodiments, but shall not limit the application of such issued claims to processes and structures accomplishing any or all of the above advantages.

[0053] Additionally, the section headings herein are provided for consistency with the suggestions under 37 C.F.R. 1.77 or otherwise to provide organizational cues. These headings shall not limit or characterize the invention(s) set out in any claims that may issue from this disclosure. Specifically, and by way of example, although the headings refer to a “Technical Field,” the claims should not be limited by the language chosen under this heading to describe the so-called field. Further, a description of a technology as background information is not to be construed as an admission that certain technology is prior art to any embodiment(s) in this disclosure. Neither is the “Summary” to

be considered as a characterization of the embodiment(s) set forth in issued claims. Furthermore, any reference in this disclosure to “invention” in the singular should not be used to argue that there is only a single point of novelty in this disclosure. Multiple embodiments may be set forth according to the limitations of the multiple claims issuing from this disclosure, and such claims accordingly define the embodiment(s), and their equivalents, that are protected thereby. In all instances, the scope of such claims shall be considered on their own merits in light of this disclosure, but should not be constrained by the headings set forth herein.

[0054] Moreover, the Abstract is provided to comply with 37 C.F.R. § 1.72(b), requiring an abstract that will allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

[0055] Any and all publications, patents, and patent applications cited in this disclosure are herein incorporated by reference as if each were specifically and individually indicated to be incorporated by reference and set forth in its entirety herein.

Claims

1. A flexible high pressure fluid cleaning lance tractor drive apparatus, comprising: a housing comprising: a front wall, a rear wall, and a top cover coupled to top ends of the front and rear walls, a bottom cover removably coupled to bottom ends of the front and rear walls, and a motor-side cover coupled to a side edge of at least one of the front wall, rear wall, and top cover; an upper belt drive unit and a lower belt drive unit both positioned within the housing, the upper and lower belt drive units configured to drive movement of at least one lance hose positioned between the upper and lower belt drive units; an actuator coupled between the top cover and the upper belt drive unit, and configured to provide vertical movement of the upper belt drive unit; a belt-side cover removably coupled to the upper belt drive unit, the belt-side cover configured to move vertically with the upper belt drive unit; and wherein the upper and lower belt drive units are slidably coupled at their front ends within a front channel formed on an interior surface of the front wall, and slidably coupled at their rear ends within a rear channel formed on an interior surface of the rear wall; and wherein one or more fasteners positioned at lower portions of the front and rear walls are configured to retain the upper and lower belt drive units within the front and rear channels.
2. The lance tractor drive apparatus of claim 1, wherein each of the upper and lower belt drive units comprises: a support wall defining a belt side and a motor side, wherein a front edge of the support wall comprises the front end slidably coupled with the front channel, and a rear edge of the support wall comprises the rear end slidably coupled with the rear channel; on the belt side of the support wall, an endless belt mounted on at least one drive roller and at least one idle roller; and on the motor side of the support wall, a motor corresponding to each of the at least one drive roller and having a drive shaft configured to turn a corresponding drive roller for moving the endless belt.
3. The lance tractor drive apparatus of claim 2, wherein a bottom edge of the support wall of the upper belt drive unit is configured to slidably engage a top edge of the support wall of the lower belt drive unit.
4. The lance tractor drive apparatus of claim 2, wherein the motor is substantially perpendicular to a rotational axis of the corresponding drive roller, and further comprising a gearbox having an input receiving the drive shaft and an output rotating the corresponding drive roller.

5. The lance tractor drive apparatus of claim 1, wherein the actuator is a pneumatic clamping mechanism.
6. The lance tractor drive apparatus of claim 1, wherein the belt-side cover comprises an upper portion and a lower portion, wherein a bottom area of the upper portion is configured to slidably engage a top area of the lower portion, and wherein only the upper portion is removably secured to the upper belt drive unit.
7. The lance tractor drive apparatus of claim 6, further comprising at least one pin coupled to the actuator and slidably passing through a corresponding at least one aperture in the support wall of the upper belt drive unit.
8. The lance tractor drive apparatus of claim 1, further comprising at least one toolless fastener securing a front end of the bottom cover to a bottom end of the front wall, and at least one toolless fastener securing a rear end of the bottom cover to a bottom end of the rear wall.
9. The lance tractor drive apparatus of claim 8, wherein the at least one toolless fastener securing the front end of the bottom cover and the at least one toolless fastener securing the rear end of the bottom cover each comprise removable pins slidably passing through corresponding apertures in the front and rear walls, and sized for clearance fit within corresponding apertures in the front and rear ends of the bottom cover.
10. The lance tractor drive apparatus of claim 8, wherein the bottom cover further comprises a bottom channel on its interior surface extending longitudinally at least partially between its front end to its rear end, and sized to removably receive a bottom edge of the lower belt drive unit.
11. The lance tractor drive apparatus of claim 1, further comprising a removable toolless fastener passing through a bottom end of the front wall at a terminal end of the front channel, and a removable toolless fastener passing through a bottom end of the rear wall at a terminal end of the rear channel, the removable toolless fasteners positioned in the front and rear walls to prevent sliding withdrawal of the upper and lower belt drive units from within the front and rear channels.
12. The lance tractor drive apparatus of claim 1, further comprising a lance hose guide entrance manifold coupled to an exterior of the rear wall and configured to position the at least one lance hose between the upper and lower belt drive units, and a lance hose guide exit manifold coupled to an exterior of the front wall and configured to position the at least one lance hose when exiting from the housing.
13. A flexible high pressure fluid cleaning lance tractor drive apparatus, comprising: a housing comprising: a front wall, a rear wall, and a top cover coupled to top ends of the front and rear walls, a bottom cover removably coupled to bottom ends of the front and rear walls, and a motor-side cover coupled to a side edge of at least one of the front wall, rear wall, and top cover; upper and lower belt drive units vertically positioned within the housing, and configured to drive movement of at least one lance hose positioned within the housing between the upper and lower belt drive units, wherein each belt drive unit comprises: a support wall defining a belt side and a motor side, wherein a front edge of the support wall is slidably coupled with a front channel on an interior of the front wall, and a rear edge of the support wall is slidably coupled with a rear channel on an interior of the rear wall, on the belt side of the support wall, an endless belt mounted on at least one drive roller and at least one idle roller, and on the motor side of the support wall, a motor corresponding to each of the at least one drive roller and having a drive shaft configured to turn a corresponding drive roller for moving the endless belt; an actuator coupled between the top cover and the upper belt drive unit, and configured to provide vertical movement of the upper belt drive unit; and a belt-side cover removably coupled to the upper belt drive unit, the belt-side cover configured to move vertically with the upper belt drive unit; wherein a bottom area of the support wall of the upper belt drive unit is configured to slidably engage a top area of the support wall of the lower belt drive unit; and wherein the bottom cover comprises a bottom channel on its interior surface extending longitudinally at least partially between its front end to its rear end, and sized to removably receive the bottom edge of the support wall of the lower belt drive unit to provide

structural support therefor.

14. The lance tractor drive apparatus of claim 13, further comprising at least one toolless fastener coupled to the actuator and slidably passing through a corresponding at least one aperture in the support wall of the upper belt drive unit, said pin configured to receive a removable tool to secure the belt-side cover to the upper belt drive unit.

15. The lance tractor drive apparatus of claim 13, further comprising at least one toolless fastener securing a front end of the bottom cover to a bottom end of the front wall, and at least one toolless fastener securing a rear end of the bottom cover to a bottom end of the rear wall.

16. The lance tractor drive apparatus of claim 13, wherein each motor is substantially perpendicular to a rotational axis of the corresponding drive roller, and further comprising a gearbox having an input receiving the drive shaft and an output rotating the corresponding drive roller.

17. The lance tractor drive apparatus of claim 13, wherein the actuator is a pneumatic clamping mechanism.

18. The lance tractor drive apparatus of claim 13, wherein the belt-side cover comprises an upper portion and a lower portion, wherein a bottom area of the upper portion is configured to slidably engage a top area of the lower portion, and wherein the upper portion is removably secured to the upper belt drive unit.

19. The lance tractor drive apparatus of claim 13, wherein at least one toolless fastener securing the front end of the bottom cover and the at least one toolless fastener securing the rear end of the bottom cover each comprise removable pins slidably passing through corresponding apertures in the front and rear walls, and sized for fit within corresponding apertures in the front and rear ends of bottom cover.

20. The lance tractor drive apparatus of claim 13, further comprising a removable toolless fastener passing through a bottom end of the front wall at a terminal end of the front channel, and a removable toolless fastener passing through a bottom end of the rear wall at a terminal end of the rear channel, the removable toolless fasteners positioned in the front and rear walls to prevent sliding withdrawal of the upper and lower belt drive units from within the front and rear channels.
